NYU SHANGHAI · ENGLISH FOR ACADEMIC PURPOSES

14 Habits of a Systems Thinker

These cards come from the **Habits of a Systems Thinker** developed by the Waters Center for Systems Thinking. Each card gives the habit, a plain-English explanation, a question to ask yourself, and a phrase you can use in seminar.

1. **Week 1:** sort the cards into three piles: *I already do this*, *I sometimes do this*, *This is hard for me*.
2. Choose **two habits you already use** and **one you find hardest**. Record them on the last card.
3. Keep the cards in your folder. You will come back to your three habits at **midterm** and in your **final reflection**.

The **Week** tag shows where each habit appears most in the course.

PRINTING
A4, actual size (100%), single-sided. Card stock works best. Cut along the dashed lines.



A SYSTEMS THINKER…

Makes meaningful connections within and between systems

Look for how things link together, and notice when two very different situations work in the same way.

ASK YOURSELF
What else works like this?

SAY IT IN SEMINAR
"This reminds me of… because both…"



A SYSTEMS THINKER…

Seeks to understand the big picture

Step back from the details to see the whole system, its purpose, and its edges, before you zoom back in.

ASK YOURSELF
If I zoomed out, what would I see that I am missing now?

SAY IT IN SEMINAR
"If we step back for a moment, the bigger issue might be…"



A SYSTEMS THINKER…

Changes perspectives to increase understanding

Look at the same situation through other people's eyes, especially people who disagree with you.

ASK YOURSELF
How would someone in a different role describe this problem?

SAY IT IN SEMINAR
"From a ___'s point of view, this might look like…"

Week 2



A SYSTEMS THINKER…

Considers how mental models affect current reality and the future

Notice the beliefs and assumptions behind what people do, including your own.

ASK YOURSELF
What belief would make this behavior make sense?

SAY IT IN SEMINAR
"I think the assumption underneath this is…"

Habits © Waters Center for Systems Thinking

Week 5



A SYSTEMS THINKER…

Observes how elements within systems change over time, generating patterns and trends

Look at change over time, not single moments. One data point is an event; many make a pattern.

ASK YOURSELF
What would a graph of this look like over a year?

SAY IT IN SEMINAR
"Over the last few years, the trend has been…"

Habits © Waters Center for Systems Thinking

Week 6



A SYSTEMS THINKER…

Surfaces and tests assumptions

Say your assumptions out loud, then check them against evidence and other views.

ASK YOURSELF
How do I know this is true? What would change my mind?

SAY IT IN SEMINAR
"What if we assumed the opposite?"

Habits © Waters Center for Systems Thinking

Week 3



A SYSTEMS THINKER…

Recognizes that a system's structure generates its behavior

When the same problem keeps happening with different people, look at the setup, not only at the individuals.

ASK YOURSELF
If everyone here were replaced, would the problem continue?

SAY IT IN SEMINAR
"This might be less about individuals and more about how the system is set up."

Habits © Waters Center for Systems Thinking



A SYSTEMS THINKER…

Identifies the circular nature of complex cause and effect relationships

Effects often loop back and become causes. Look for reinforcing and balancing loops.

ASK YOURSELF

How does the result feed back into the cause?

SAY IT IN SEMINAR

"That creates a loop: the more..., the more..."

Habits © Waters Center for Systems Thinking



A SYSTEMS THINKER…

Recognizes the impact of time delays when exploring cause and effect relationships

Results often arrive later than actions. Delays can make good choices look bad and bad choices look good.

ASK YOURSELF

When will we actually see the effect of this action?

SAY IT IN SEMINAR

"We might not see the results until..."

Habits © Waters Center for Systems Thinking



A SYSTEMS THINKER…

Considers short-term, long-term and unintended consequences of actions

Ask what happens next, and after that. Quick fixes can create new problems somewhere else.

ASK YOURSELF

Who else is affected, and what happens in five years?

SAY IT IN SEMINAR

"In the short term this helps, but in the long term..."

Habits © Waters Center for Systems Thinking



A SYSTEMS THINKER…

Considers an issue fully and resists the urge to come to a quick conclusion

Slow down. Gather information and listen before you decide.

ASK YOURSELF

What do I still not know?

SAY IT IN SEMINAR

"Before we decide, could we look at...?"

Habits © Waters Center for Systems Thinking



A SYSTEMS THINKER…

Pays attention to accumulations and their rates of change

Notice what builds up or drains away over time, such as savings, trust, stress, or carbon, and how fast it is changing.

ASK YOURSELF

What is filling this up, and what is draining it?

SAY IT IN SEMINAR

"Even though it is slowing down, it is still building up."



A SYSTEMS THINKER…

Uses understanding of system structure to identify possible leverage actions

Look for the place where a small, well-chosen change could make a big difference.

ASK YOURSELF

Where would a small change have the biggest effect?

SAY IT IN SEMINAR

"The real leverage point here might be…"



A SYSTEMS THINKER…

Checks results and changes actions if needed: "successive approximation"

Try, check, adjust, and try again. Improvement comes in rounds.

ASK YOURSELF

What did I learn from my last attempt, and what will I change?

SAY IT IN SEMINAR

"Based on the feedback, next time I will…"

My three habits

A habit I already use

Example:

Another habit I already use

Example:

The habit I find hardest

Example:

Midterm check (Week 8): what has changed?

Upload your Week 1 answers to Student Folder → Week 1 by Sun Jan 24, 23:59.

Name: _____